Task 1: Window Functions & Advanced Analytics

Task A (Ranking): Write a query to assign a rank to employees with highest salary based on their department

```
select employee_fname,salary,dept_id,
rank() over(
        partition by dept_id
        order by salary desc) as rank_based_on_salary
from employee;
```

Explanation:

This query assigns a rank to employees within each department based on their salary in descending order. The RANK() function with PARTITION BY dept_id ensures that ranking restarts for each department, so the highest-paid employee in each department gets rank 1.

employee 1

select employee_fname,salary,dept_id, rank() over(| Enter a SQL expression to filter results (use Ctrl+Spac

	employee_fname	salary	dept_id	rank_based_on_salary
1	Zara	47,000	101	1
2	Varun	47,000	101	1
3	Aarav	46,000	101	3
4	Amit	45,000	101	4
5	Adarsh	56,000	102	1
6	Bhavya	54,000	102	2
7	Riya	52,000	102	3
8	Sneha	51,000	102	4
9	Binit	65,000	103	1
10	Abhishek	62,000	103	2
11	Chirag	61,000	103	3
12	Rahul	48,000	103	4
13	Pallavi	54,000	104	1
14	Disha	53,000	104	2
15	Charul	52,000	104	3
16	Neha	50,000	104	4
17	Dev	74,000	105	1
18	Harsh	72,000	105	2
19	Eshan	67,000	105	3
20	Karan	65,000	105	4
21	Ekta	49,000	106	1

Task B (Running Total): specialized task to calculate the cumulative sum of salary each dept.

```
select distinct dept_name, dept_id, total_sal
from (
   select d.dept_name,
         d.dept_id,
         sum(e.salary) over (partition by e.dept_id) as total_sal
   from department d
   join employee e
     on e.dept_id = d.dept_id
) t;
```

Explanation:

This query calculates the cumulative salary for each department using the SUM() window function with PARTITION BY e.dept_id. It joins employee and department tables to show each department's total salary alongside its name and ID.

department 1

select distinct dept_name, dept_id, total_sal from (Enter a SQL exp

Grid Text

	dept_name	dept_id	total_sal
1	Marketing	104	209,000
2	Directors	123	585,000
3	UI/UX Design	116	229,000
4	Training and Developm	125	193,000
5	Research and Develop	117	318,000
6	Quality Assurance	109	203,000
7	Project Management	115	435,000
8	Finance	102	213,000
9	Technical Support	120	198,000
10	Product Management	114	426,000
11	Cyber Security	111	323,000
12	Business Analytics	113	251,000
13	Sales	103	236,000
14	Networking	119	309,000
15	Administration	106	178,000
16	Team Leads	121	302,000
17	Cloud Engineering	118	315,000

Task C (Lead/Lag): Display each employee's salary along side his/her previous employee's salary.

```
select employee_fname,salary,
lag(salary) over() as prev_salary,
lead(salary) over() as next_salary
from employee;
```

Explanation:

This query displays each employee's salary along with the salary of the previous and next employee using LAG() and LEAD() window functions. It helps compare an employee's salary to their neighboring records in the result set.

select employee_fname,salary, lag(salary) over() as | Enter a SQL expression to filter results (use C

Grid | Text

	employee_fname	salary	prev_salary	next_salary
1	Amit	45,000	[NULL]	52,000
2	Riya	52,000	45,000	48,000
3	Rahul	48,000	52,000	50,000
4	Neha	50,000	48,000	65,000
5	Karan	65,000	50,000	42,000
6	Priya	42,000	65,000	55,000
7	Suresh	55,000	42,000	70,000
8	Ananya	70,000	55,000	48,000
9	Vikram	48,000	70,000	72,000
10	Pooja	72,000	48,000	80,000
11	Arjun	80,000	72,000	90,000
12	Mehul	90,000	80,000	60,000
13	Tanvi	60,000	90,000	95,000
14	Rakesh	95,000	60,000	98,000
15	Nitin	98,000	95,000	55,000
16	Sonal	55,000	98,000	75,000
17	Mohit	75,000	55,000	82,000
18	Ayesha	82,000	75,000	68,000
19	Rohan	68,000	82,000	46,000